

QUESTION 1: ESSAY

For $A \rightarrow B$ reaction (Pressure = P_A) occurring on a catalyst surface, the combined rate of adsorption, desorption and surface reaction steps is given by $-r_A = (P_A^3 - P_B^2/5)/(1 + P_A + P_B)^2$. The mass transfer coefficient for A is given by k_c . When the system is agitated, the overall rate of the process also changes, implying that this is a mass-transfer limited process and the surface reaction is occurring very fast. Find the overall rate of reaction (at bulk pressures = P_A and P_B). Assume no internal diffusion limitation.

Given Answer: mass transfer = rate of reaction at steady state.

$$-r_A = K_c (C_A - C_B) = (P_A^3 - P_B^2/5)/(1 + P_A + P_B)^2$$

Score: 25

QUESTION 2: ESSAY

Why is it important for chemisorption to occur for catalysis? Why not physical adsorption?

Given Answer: Chemisorption affects reaction rates for chemical reactions. This is highly exothermic, and electronic structure is perturbed. This is the actual reaction taking place. Physical absorption is like condensation and it's a weak interaction. only slightly exothermic.

QUESTION 3: ESSAY

List the seven steps of a catalytic reaction.

Given Answer:

- 0 - bulk flow
- 1 - mass transfer to the surface of the catalyst
- 2- pore diffusion into the catalyst
- 3 - adsorption
- 4- reaction on catalyst site
- 5 - desorption of products
- 6 - pore diffusion out of catalyst
- 7 - mass transfer out of bulk
- 8- bulk moves forward.



Correct Answer: 

Bulk Flow
Mass-Transfer in
Internal Diffusion in
Adsorption
Reaction

Desorption
Diffusion out
Mass-Transfer out
Bulk Flow out

Score: 25

QUESTION 4: ESSAY

What is the meaning of rate-limiting step in a catalytic reaction. Explain in terms of rate constants and resistances.

Given
Answer: A rate limiting step in reaction is the slowest part of a reaction. This is what defines the rate. For example, if adsorption is rate limiting step, $k_{ad} \ll k_s, k_d$. and $r_d/k_d = r_s/k_s = 0$. in this case of resistances, $1/k_{ad}$ will be much bigger than other resistances thus making it the limiting step.

Score: 25